

Challenge: Perimeter Patrol

Points: 200 • Estimated Time: 45 Minutes

Challenge Scope

The robot must follow the outer border of a designated 2-meter square taped boundary for two complete loops.

Instructions & Engineering Hints

- **Your Mission:** Program your robot to successfully navigate and patrol the outer perimeter of a 2-meter square taped boundary for exactly two complete loops without losing the track or cutting corners.
- **Sensor Strategy:** Consider how your robot will detect the boundary line. Will you use a single light/color sensor, or would a dual-sensor configuration help keep the robot aligned parallel to the tape? Think about sensor placement relative to the wheels—mounting sensors closer to the pivot point can improve reaction time.
- **Navigation Logic:** The square has four distinct 90-degree turns. How will your robot recognize when it has reached a corner versus following a straight line? You may need to incorporate proportional line-following logic or a state-machine approach that counts completed corners to track your two full loops.
- **Structural Considerations:** Ensure your robot has a wide enough wheel base to prevent tipping during sharp pivot turns, and make sure your cables are securely routed so they don't interfere with your ground sensors.